New model evaluation and multi-strategy backtest

Updated 2026-10-14. Results were generated independently from the four model delivery packages and basic data.zip provided this time, without overwriting the previous version.

Unified six-year rolling training, refits every 26 weeks, 5/7/11/45 factors. 15 of the original 16 configurations are valid: four each for RF, regression XGBoost and Ranker, and three for Ridge; Ridge45 did not pass the original Development gate. One four-model consensus was also built for each of 5/7/11 factors, giving 18 series in total.

Each series runs 15 strategies, 14 time windows and one-way 0/10/25bps: 11,340 main backtests in total, plus 972 rebalance-phase tests.

Main finding: in the Final Test at 10bps, among the original models' equal-weight Top5, the 5-factor Ranker performed best, with net annualised return 22.64% and maximum drawdown 11.35%. Rankings change when the strategy changes, so model quality cannot be judged from a single highest-return combination.

Key source limitation: Ranker and Ridge essentially reproduce the delivered tables; RF return metrics agree while some correlation metrics differ slightly; regression XGBoost shows clear inconsistencies, and running the original script directly also confirmed the difference for 5 factors. XGB in this report means the predictions recovered from the delivered script this time, not the unknown original predictions implied by the delivered CSV.

Data, timing and prediction recovery

The SHA-256 of the five weekly factor files is identical: the raw panel has 27,651 rows, 66 stocks and 522 dates, covering 2016-09-02 to 2026-08-28. The input has 45 factors. The 5 factors are revenue growth, operating profitability, cash flow / assets, debt / assets and asset growth; the 7 factors are the original priority set; the 11 factors are the 7 factors plus 4 macro factors.

The first 6 years are used for initial model training. Full-history backtest signals run from 2022-09-02 to 2026-08-21, 208 weeks in total; the last return is realised on 2026-08-28. The ten years of raw data are not cropped; the first 6 years simply have no out-of-sample predictions under this design.

Development: 2022-09-02 to 2024-08-30, 105 weeks; Validation: 2024-09-06 to 2025-08-29, 52 weeks; Final Test: 2025-09-05 to 2026-08-21, 51 weeks. Every valid original model has 3,304 same-date, same-stock predictions in the Final Test.

RF is recovered with the frozen parameters from the delivered table; Ridge runs the original 6-alpha search; XGB runs the original 12-candidate search; Ranker runs the original 36-candidate label and parameter search and freezes it. Ridge and RF select on 104 weeks of realised labels; XGB and Ranker keep the original 105-week convention, which includes the end-of-Development label crossing into the start of Validation.

The original packages contain no per-stock predictions, so the backtest has to recover the model outputs. The item-by-item reconciliation between the delivered summaries and the reruns is saved in source_metric_reconciliation.csv.

Reconciliation of delivered results with this recovery

The table gives the largest absolute difference across factor sets and stages for each family. Differences in annualised return are in percentage points; e.g. 52.03% means 52.03 percentage points.

All eight Ranker metrics agree exactly; Ridge differs only by floating-point error; RF annualised and excess returns agree, while some correlation metrics differ slightly. This backtest uses the recomputed recovered predictions.

XGB differs significantly. Rerunning the original 5-factor entry point directly gives a Final Test gross annualised return of −20.80% versus −13.58% in the delivered table; the direct run agrees with this recovery. The other factor sets differ as well. The delivered package did not keep frozen parameters or per-stock predictions, so the reason the original summary differs cannot be confirmed, and the difference cannot be attributed to a single proven cause.

The XGB results are therefore labelled separately as a script-reproduction experiment and cannot be used to claim an exact reproduction of the delivered model. Checking the original delivered results would still require the original per-stock predictions or frozen configuration.

Model	Max RankIC difference	Max annualised return difference	Max IR difference
RANKER	0.000	0.00%	0.000
RF	0.000	0.00%	0.000
RIDGE	0.000	0.00%	0.000
XGB	0.065	52.03%	3.638

Portfolio strategies and cost method

15 strategies in total. All filters, volatility and trend indicators use only data up to and including the signal date. Unfilled slots stay in cash, and cash earns 0. The positive-score strategies are only a score-threshold experiment for the Ranker and must not be read as removing stocks expected to lose money.

Stock return = next global week's adjusted close / current week's adjusted close − 1; the gross portfolio return is the weighted sum of stock returns. The buy/sell traded weight is the sum of absolute differences between target weights and last week's weights after return drift. Net return = (1 − traded weight × one-way cost) × (1 + gross return) − 1.

Every window starts from cash with net value 1; the buffer and 2/4-week strategies are replayed per window. The first purchase is charged, and the end is marked to market without forced liquidation. The benchmark is the gross equal-weight return of the model's evaluable stocks in that week, so the comparison with the net strategy is not fully symmetric.

Category	Strategy	Rule
Equal weight	Top5, Top10	Select by rank, 1/5 or 1/10 each
Positive score	Top5, Top10	prediction > 0; a Ranker score is not a profit forecast
Past-loss filter	4w, 13w × Top5/10	A stock enters only if its past return is non-negative
Rank weighted	Top5	5:4:3:2:1, unfilled slots stay in cash
Risk weighted	Low-vol, rank-risk Top5	Inverse 26-week volatility, at most 35% per stock
Buffer	Top5_Buffer10	Keep existing stocks that are still in the top 10
Low frequency	2-week, 4-week Top5	Weights drift naturally in non-rebalancing weeks
Trend	26-week market trend Top5	Buy only when the equal-weight universe index is above its 26-week average

Predictive power: Final Test

Pearson IC and Spearman RankIC are computed across stocks each week, then summarised as the mean, mean / standard deviation and the share of weeks with positive RankIC. ICIR and RankICIR are not annualised.

Regression models additionally report MAE/RMSE of log returns, with a zero-prediction RMSE benchmark in the CSV. The Ranker and the consensus output ranking scores, so no return-prediction error is computed. Predictive correlation and portfolio profitability should be judged separately.

Model	IC	RankIC	RankICIR	Positive RankIC weeks	RMSE
XGB-R6-F7	0.039	0.057	0.349	70.59%	0.059
RANKER-R6-F5	0.048	0.050	0.361	66.67%	—
RF-R6-F7	0.045	0.045	0.307	64.71%	0.059
RIDGE-R6-F7	0.030	0.042	0.230	64.71%	0.059
RIDGE-R6-F11	0.028	0.037	0.208	66.67%	0.059
XGB-R6-F45	0.012	0.031	0.191	56.86%	0.059
RANKER-R6-F7	0.032	0.030	0.188	56.86%	—
XGB-R6-F11	0.024	0.029	0.217	56.86%	0.060
RIDGE-R6-F5	0.033	0.027	0.208	58.82%	0.059
RF-R6-F11	0.040	0.026	0.188	52.94%	0.059
RANKER-R6-F45	0.004	0.024	0.152	49.02%	—
RF-R6-F5	0.010	0.021	0.124	49.02%	0.060
XGB-R6-F5	-0.009	0.015	0.086	50.98%	0.060
RANKER-R6-F11	0.007	0.012	0.080	49.02%	—
RF-R6-F45	0.014	0.008	0.046	47.06%	0.059

Equal-weight Top5: the common baseline across models

Models are compared after unifying strategy and costs; all figures below use one-way 10bps. The 208-week full history includes the Development tuning period and should not be called an unseen test throughout.

In the Final Test, only 4 of the 15 valid original configurations have a positive equal-weight Top5 net annualised return: Ranker5, Ranker7, Ranker11 and the recovered XGB11. RF and Ridge equal-weight Top5 are negative in this stage.

Model	Full-history net ann.	Validation net ann.	Final Test net ann.	Final Test max drawdown
RANKER-R6-F11	3.92%	1.01%	1.28%	-19.98%
RANKER-R6-F45	-9.85%	-12.19%	-14.64%	-27.87%
RANKER-R6-F5	11.93%	-18.10%	22.64%	-11.35%
RANKER-R6-F7	11.58%	11.86%	3.25%	-16.95%
RF-R6-F11	-0.12%	-10.23%	-3.99%	-23.70%
RF-R6-F45	-0.35%	8.79%	-15.50%	-21.33%
RF-R6-F5	3.42%	6.75%	-12.52%	-32.12%
RF-R6-F7	-6.93%	-10.26%	-27.96%	-32.94%
RIDGE-R6-F11	-0.70%	-7.45%	-5.65%	-26.26%
RIDGE-R6-F5	-4.08%	18.11%	-15.10%	-20.55%
RIDGE-R6-F7	-3.65%	-11.43%	-13.82%	-21.71%
XGB-R6-F11	2.65%	-10.31%	5.89%	-20.02%
XGB-R6-F45	-7.63%	-26.97%	-11.51%	-26.04%
XGB-R6-F5	-1.92%	15.07%	-21.53%	-42.53%
XGB-R6-F7	-0.72%	6.93%	-4.00%	-28.80%

Multi-strategy results: best Final Test combination per model

This table is a purely ex-post description: comparing 15 strategies within the same Final Test and picking the best introduces multiple-comparison and selection bias. These strategies were not selected on independent unseen data as deployable configurations.

The recovered XGB5 13-week loss-filter Top5 reaches 37.70%, but has the source-reproduction difference; the reproducible Ranker5 buffer Top5 reaches 37.38% with a maximum drawdown of 10.05%. RF11 with 4-week rebalancing reaches 35.94% and needs further checking for phase dependence.

Model	Best Final Test strategy	Net ann.	Max drawdown	Sharpe
XGB-R6-F5	Top5_NoLoss13w	37.70%	-21.44%	1.395
RANKER-R6-F5	Top5_Buffer10	37.38%	-10.05%	1.598
RF-R6-F11	Top5_Rebalance4w	35.94%	-18.46%	1.431
XGB-R6-F7	Top5_NoLoss13w	30.85%	-21.78%	1.152
RANKER-R6-F45	Top5_Rebalance4w	29.90%	-20.86%	1.120
RIDGE-R6-F5	Top5_NoLoss13w	29.59%	-13.71%	1.158
RF-R6-F5	Top5_NoLoss13w	26.48%	-18.49%	0.995
RANKER-R6-F7	Top5_Rebalance4w	25.83%	-18.30%	0.915
RIDGE-R6-F7	Top5_NoLoss13w	19.42%	-17.01%	0.846
RIDGE-R6-F11	Top5_NoLoss13w	18.34%	-18.19%	0.818
RF-R6-F45	Top5_Rebalance2w	15.71%	-13.16%	0.784
RF-R6-F7	Top10_NoLoss13w	14.93%	-22.50%	0.725
XGB-R6-F45	Top5_Rebalance4w	10.96%	-14.99%	0.555
XGB-R6-F11	Top5_RankRisk26w	10.35%	-18.24%	0.509
RANKER-R6-F11	Top5_Rebalance4w	8.51%	-24.10%	0.412

Coverage of strategy improvements

The counts cover the 15 valid original models at one-way 10bps; the three derived consensus series are not counted again as independent original models. An improvement means better than equal-weight Top5 and may only mean a smaller loss.

Strategies should be judged on breadth of improvement, the validation period, the full history, transaction costs and phase together, not only on the single best model.

Strategy	Profitable in test	Better test return	Better test drawdown	Profitable over full history
Top10	1	10	11	5
Top10_NoLoss13w	11	11	11	8
Top10_NoLoss4w	6	8	12	9
Top10_PositiveScore	2	12	14	7
Top5	4	0	0	5
Top5_Buffer10	4	8	8	4
Top5_LowVol26w	7	12	11	4
Top5_NoLoss13w	9	9	10	9
Top5_NoLoss4w	3	8	10	8
Top5_PositiveScore	3	7	4	6
Top5_RankRisk26w	7	12	10	4
Top5_RankWeighted	4	5	2	6
Top5_Rebalance2w	10	11	12	9
Top5_Rebalance4w	12	12	10	7
Top5_Trend26w	3	6	11	2

Transaction costs and comparison across periods

The table lists representative combinations, keeping the same holding path at 0/10/25bps. Low frequency and buffering can reduce trading frequency, but return improvements may also come from the timing of switches and the holding path.

The best original-model combination over the full history is Ranker7 with 2-week rebalancing, 30.37% annualised at 10bps; its Final Test result and starting-week sensitivity must be read together.

Model	Strategy	Test 0bps	Test 10bps	Test 25bps	Full history 10bps
RANKER-R6-F5	Top5_Buffer10	38.25%	37.38%	36.07%	15.38%
RANKER-R6-F7	Top5_Rebalance2w	18.10%	14.25%	8.70%	30.37%
RF-R6-F11	Top5_Rebalance4w	39.06%	35.94%	31.38%	-2.17%
XGB-R6-F5	Top5_NoLoss13w	41.20%	37.70%	32.60%	5.36%

Rebalance starting-week sensitivity

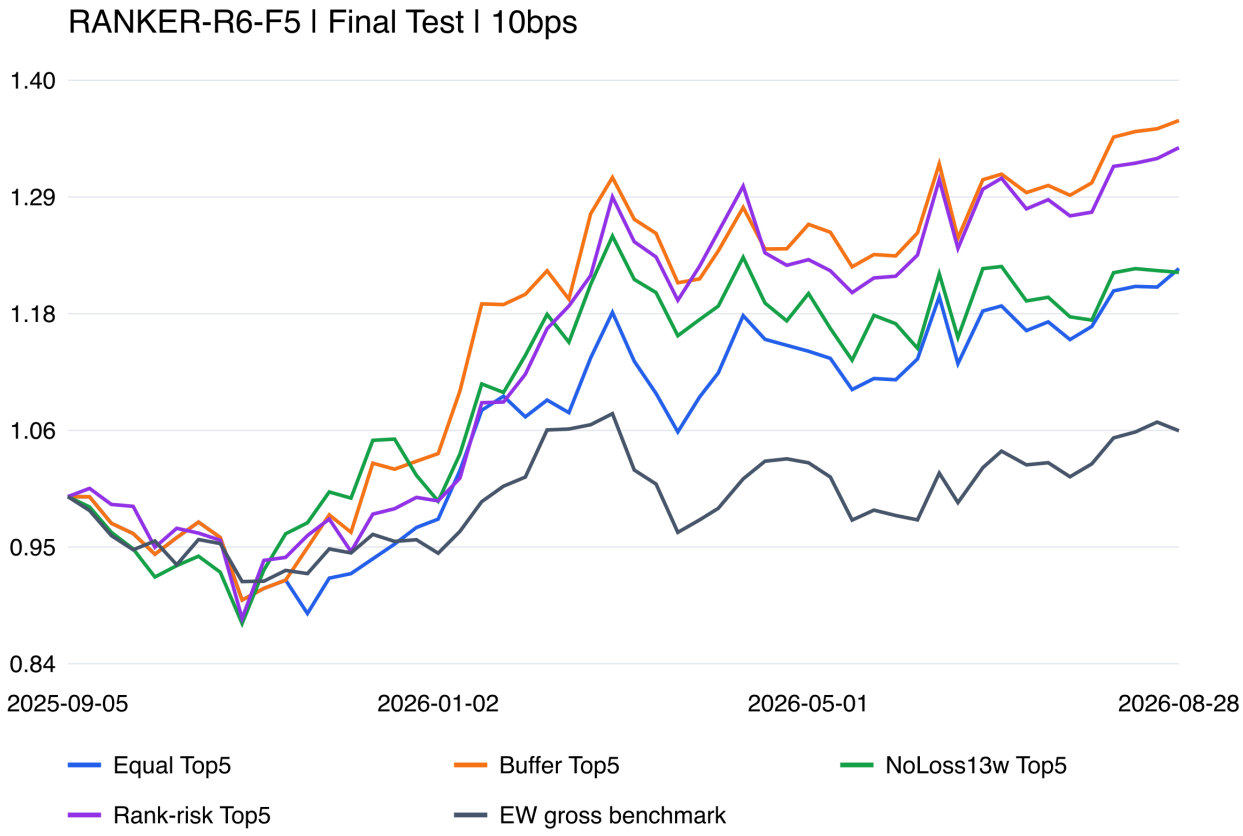
2-week rebalancing tests an initial wait of 0/1 week, and 4-week rebalancing tests 0/1/2/3 weeks. Cash is held during the wait, and every variant keeps the whole window's weeks in the denominator. Phase 0 matches the main backtest; the best phase must not be picked to replace the predefined result.

A large gap between the minimum and maximum return in the table means the good result depends on the rebalancing start.

Model	Strategy	Main phase net ann.	Worst phase	Best phase
RF-R6-F11	Top5_Rebalance4w	35.94%	-17.07%	35.94%
RANKER-R6-F7	Top5_Rebalance2w	14.25%	14.25%	34.07%
RANKER-R6-F5	Top5_Rebalance4w	23.49%	11.67%	28.15%
RANKER-R6-F45	Top5_Rebalance4w	29.90%	7.95%	29.90%

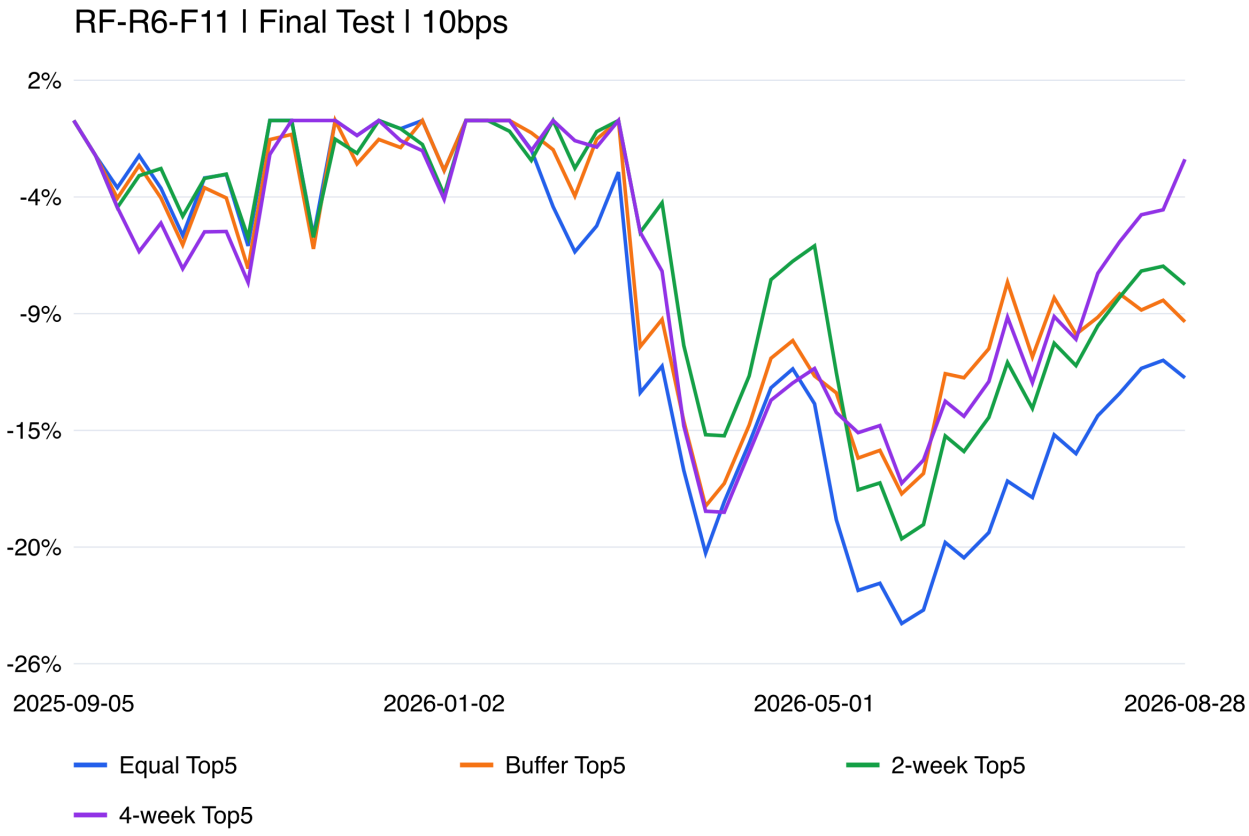
Ranker5: Final Test net value of representative strategies

Equal weight, buffer, 13-week loss filter and rank-risk weighting are selected for comparison in the text; all 15 strategies are in the interactive dashboard. All curves start at net value 1 at one-way 10bps; the grey benchmark is the gross equal-weight return of the same universe.



RF11: Final Test drawdown comparison

Drawdown is the fall of the current net value from its previous peak, including the initial net value of 1. The highest-return strategy does not necessarily have the smallest drawdown, and low-frequency results must be read with the phase test on the previous page.



Four-model consensus

For each of the 5/7/11 factor sets, the four families are fixed and their weekly prediction percentiles within their own universe are equal-weight averaged, including only stocks predicted by all four; the combined score is rounded to 12 decimals to handle numerical ties. Ridge45 was not admitted, so no 45-factor four-model consensus is built.

The consensus is an additional derived prediction that runs all 15 strategies and does not replace any model. The table shows equal-weight Top5 at 10bps; combining models does not guarantee an improvement.

Consensus	Full-history net ann.	Validation net ann.	Final Test net ann.
CONSENSUS-R6-F11	-2.86%	-19.21%	-18.69%
CONSENSUS-R6-F5	-1.64%	-6.23%	-3.77%
CONSENSUS-R6-F7	6.65%	9.43%	-1.03%

Verification, research limitations and reproduction

The independent verification passed 3,503,939 checks with a maximum absolute error of 2.62e-14. It checks direct price returns, every model's per-window strategy holdings, drift and buy/sell costs, the metrics at all three cost levels, phase 0 and the consensus scores.

Risks and limitations: same-close signals with same-close entry are a research assumption; next-open execution, slippage, impact costs, liquidity and capacity are not verified. The risk-free rate and cash return are 0, and the benchmark is gross of costs.

The evaluable universe depends on whether a next-week price exists, which creates an ex-post evaluability selection bias. When a held stock lacks a next-period price it exits at the currently available price with a sell cost; this is a research treatment and cannot be regarded as a real delisting fill. Historical factor and universe screening may also have used information from the whole period.

The full history includes the tuning stage, the Final Test has been viewed in several rounds of project research, and the new strategies are also ex-post exploration; a new time period or nested time validation is needed to assess generalisation. Regression XGB has the additional source limitation that the delivered metrics cannot be reproduced.

Reproduction order (original workflow): run replay_models.py in this folder, run the Ridge and Ranker scripts in the packages with output to the replay folder, then run evaluate_latest.py, verify_latest.py and build_report.py. In this package the reproducible version is research/06_evaluation_backtest (evaluate.py and verify.py).